Modern Reinforcement Learning: From Agents to LLMs

Post-Training LLMs: KL-Regularized Policy Optimization
SFT → PPO → DPO → GRPO

Lecture 8

Keywords: LLM as policy, SFT, Preferences, Policy Optimization, PPO, DPO, GRPO

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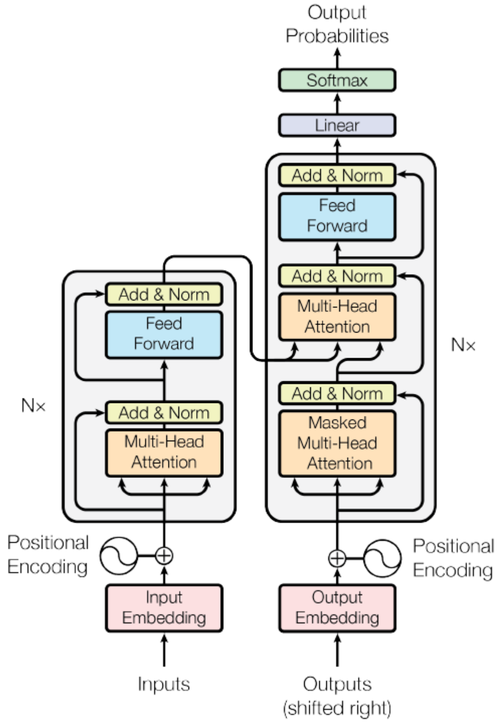
Eval form

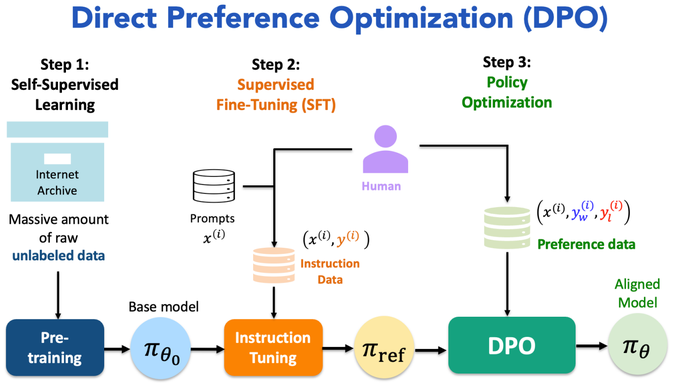
https://evasys.unibas.ch/evasys/online.php?pswd=SZUTZ

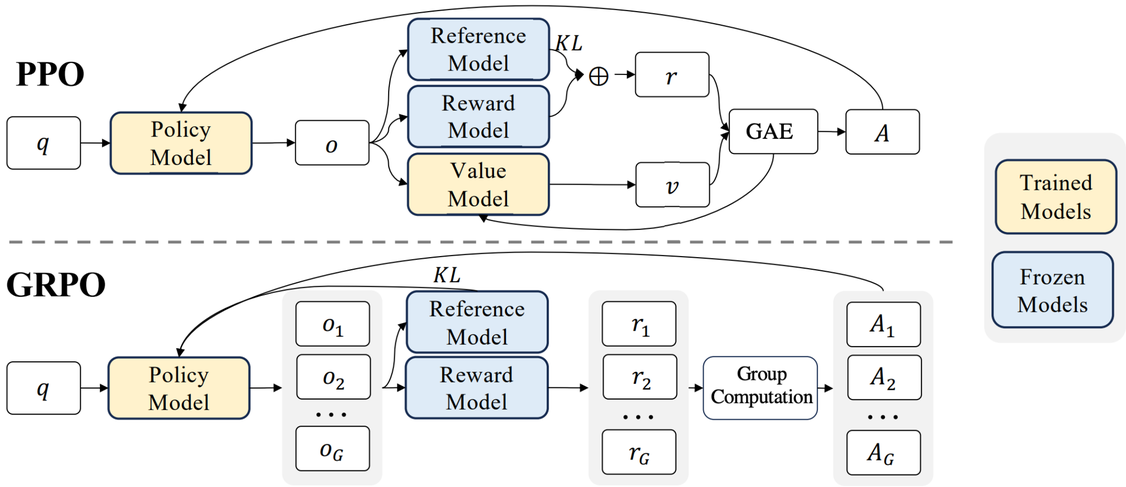
Lecture A

Today's Roadmap

1. Transformer → policy view (attention + causal mask)
2. SFT objective and why it defines π_{ref}
3. KL-regularized post-training objective
4. Closed-form optimum π^*
5. PPO → DPO → GRPO (connections)
6. Demos

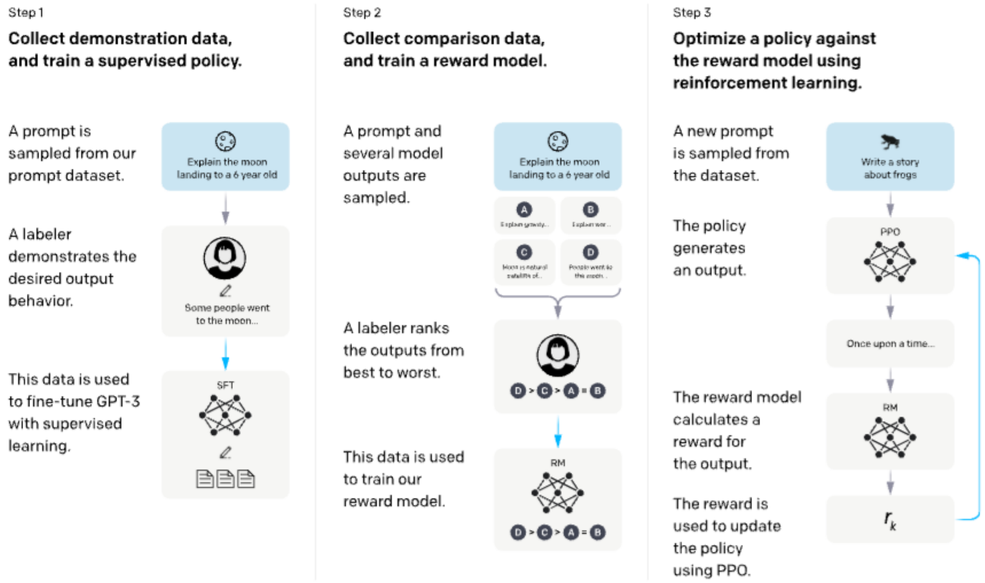






Course Map: From Preferences to Policy Optimization

Pretraining → SFT → Preference data (BT) → RM or Direct → KL-regularized optimization



- Lecture 7: preferences + Bradley–Terry + reward models
- Lecture 8A: SFT + KL-regularized objective
- Lecture 8B: PPO / DPO / GRPO + demos

LLM As a Stochastic Policy

- Prompt x, completion $y = (y_1, \ldots, y_T)$

- Autoregressive policy factorization:

$$\pi_\theta(y \mid x) = \prod_{t=1}^{T} \pi_\theta(y_t \mid x, y_{<t})$$

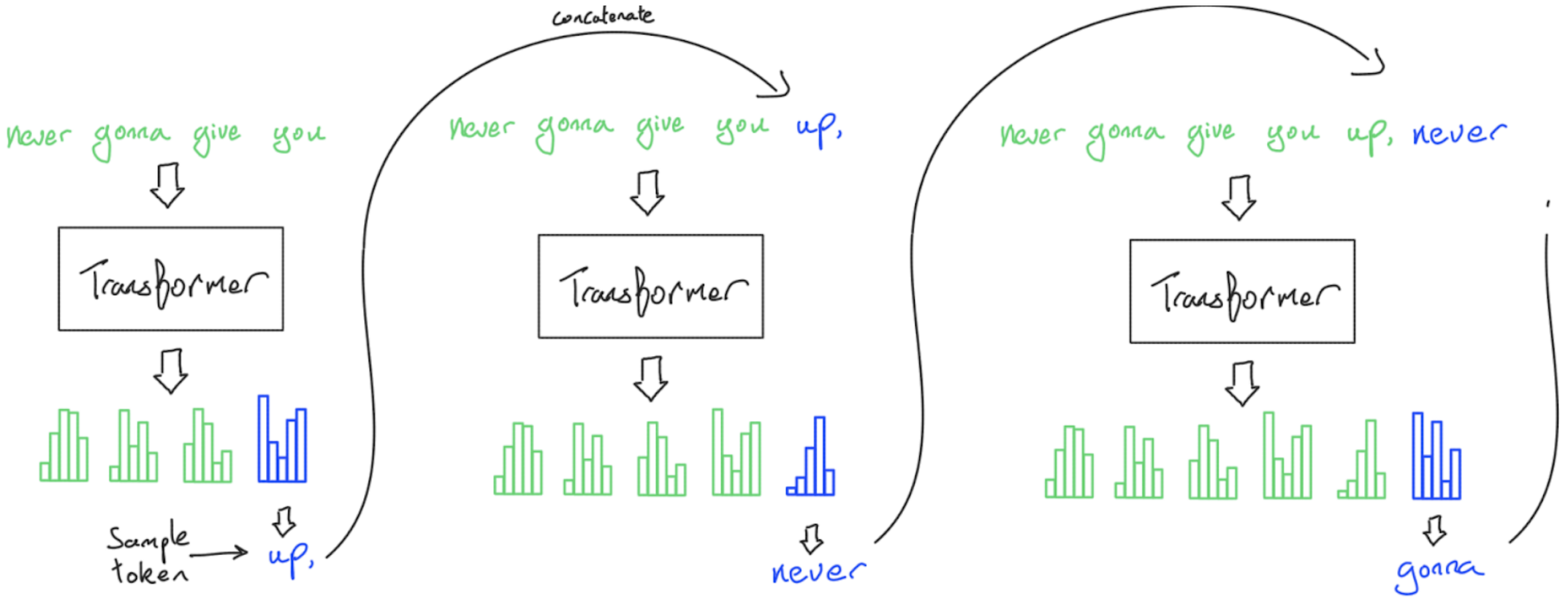
- Log-prob decomposes over tokens:

$$\log \pi_\theta(y \mid x) = \sum_{t=1}^{T} \log \pi_\theta(y_t \mid x, y_{<t})$$

📘 **In RL terms:**

State: $s_t = (x, y_{<t})$

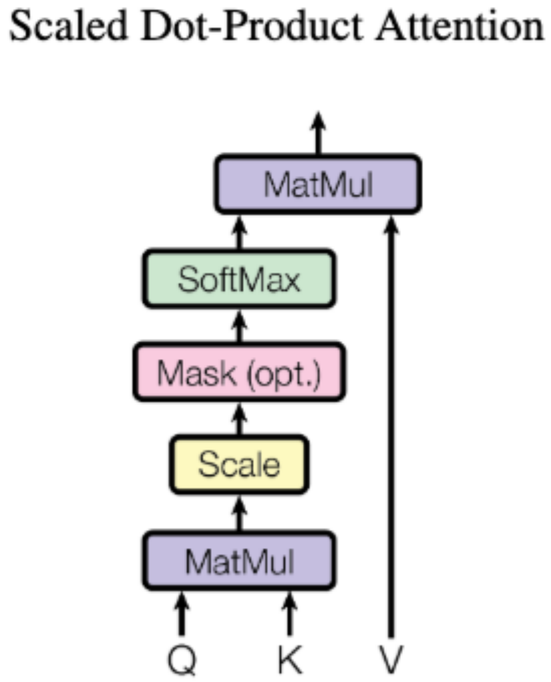
Action: $a_t = y_t$


never gonna give you
Transformer
Sample token
up,
concatenate
never gonna give you up,
Transformer
never
never gonna give you up, never
Transformer
gonna

Self-Attention As The Core Primitive

- Input hidden states $H \in \mathbb{R}^{T \times d}$
- Linear projections: $Q = HW_Q,\ K = HW_K,\ V = HW_V$
- Scaled dot-product attention:

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$



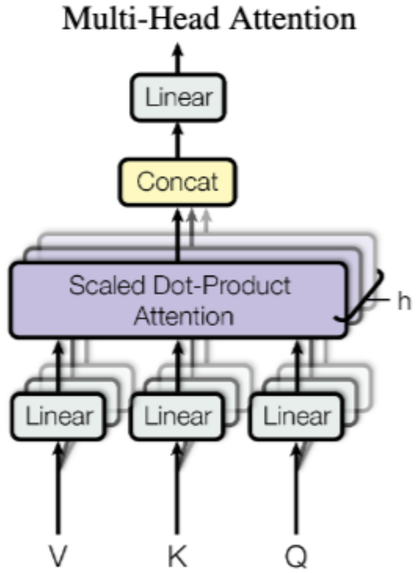
Causal Masking Gives The Policy Factorization

- **Causal (triangular) mask:** token t attends only to positions $\leq t$
- Masked attention logits:

$$\text{softmax}\left(\frac{QK^\top + M}{\sqrt{d_k}}\right), \qquad M_{t,j} = \begin{cases} 0 & j \leq t \\ -\infty & j > t \end{cases}$$

- Consequence: next-token distribution depends only on $\pi_\theta(y_t \mid x, y_{<t})$

Multi-Head Attention



- Multiple heads attend in parallel:

$$\text{head}_i = \text{Attn}(HW_Q^{(i)}, HW_K^{(i)}, HW_V^{(i)})$$

- Concatenate + project:

$$\text{MHA}(H) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)\, W_O$$

- Intuition: different heads learn different matching functions over positions/features.

Pretraining Objective (Maximum Likelihood)

- Dataset of text: (x, y) with completion tokens $y = (y_1, \ldots, y_T)$
- Teacher-forcing MLE:

$$\max_{\theta} \; \mathbb{E}_{(x,y)\sim\mathcal{D}} \left[\sum_{t=1}^{T} \log \pi_\theta(y_t \mid x, y_{<t}) \right]$$

- Equivalent: minimize cross-entropy / negative log-likelihood.

Why Pretraining Is Not Enough

- Pretraining fits **text distribution**: what appears on the internet
- Desired behavior is conditional and normative:
  - follow user intent (helpfulness)
  - avoid unsafe outputs (harmlessness)
  - be truthful / calibrated (honesty)
- Mismatch: “most likely continuation”≠“best assistant action”

Alignment Targets As Objectives

- We want behavior that is helpful, honest, and harmless (HHH)
- Model this as reward design (or multi-objective reward):

$$r(x, y) = w_H\, r_{\text{help}}(x, y) + w_T\, r_{\text{truth}}(x, y) - w_S\, r_{\text{harm}}(x, y)$$

- Or as a constraint (common in safety):

$$\max_{\theta} \; \mathbb{E}[r_{\text{help}}] \quad \text{s.t.} \quad \mathbb{E}[r_{\text{harm}}] \leq \delta$$

- Post-training methods differ in how they optimize, not in what they target

Supervised Fine-Tuning

- Data: instruction prompts x with **demonstration** responses $y^\star$
- Train by teacher forcing (next-token prediction on the demo)
- SFT produces a strong reference policy π_{ref} for later stages

$$\max_{\theta} \; \mathbb{E}_{(x,y^\star)\sim\mathcal{D}_{\text{demo}}} \left[\sum_{t=1}^{T} \log \pi_\theta\left(y_t^\star \mid x, y_{<t}^\star\right) \right]$$

📘 SFT ≈ imitate good behavior; RLHF ≈ improve it under preference/ safety signals.”

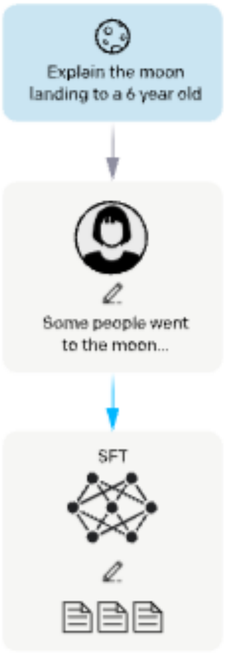
Step 1

Collect demonstration data, and train a supervised policy.

A prompt is sampled from our prompt dataset.

A labeler demonstrates the desired output behavior.

This data is used to fine-tune GPT-3 with supervised learning.



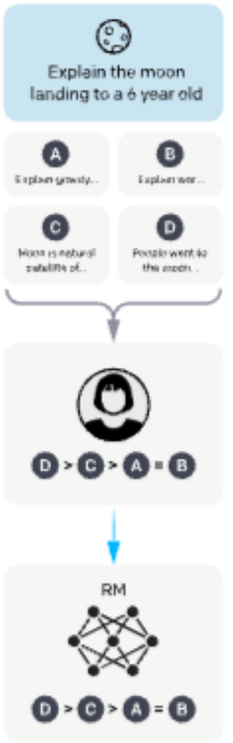
Step 2

Collect comparison data, and train a reward model.

A prompt and several model outputs are sampled.

A labeler ranks the outputs from best to worst.

This data is used to train our reward model.



Step 3

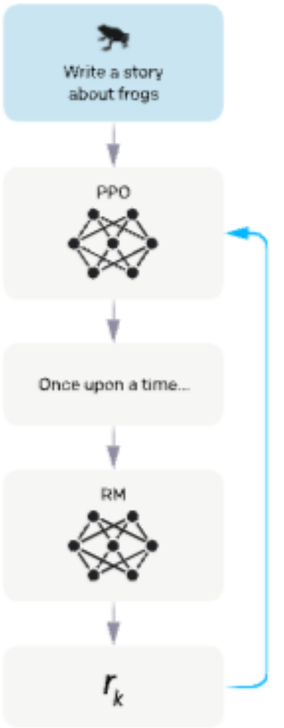
Optimize a policy against the reward model using reinforcement learning.

A new prompt is sampled from the dataset.

The policy generates an output.

The reward model calculates a reward for the output.

The reward is used to update the policy using PPO.



SFT As KL Minimization

- Let $p_{\text{demo}}(y \mid x)$ be the (unknown) demonstration distribution
- SFT maximizes $\mathbb{E}_{p_{\text{demo}}} \log \pi_\theta(y \mid x)$
- Equivalent viewpoint: fit $\pi_\theta(\cdot \mid x)$ to $p_{\text{demo}}(\cdot \mid x)$

$$\arg\max_\theta \; \mathbb{E}_{(x,y)\sim p_{\text{demo}}}\left[\log \pi_\theta(y \mid x)\right] \;=\; \arg\min_\theta \; \mathbb{E}_x\left[\mathrm{KL}\left(p_{\text{demo}}(\cdot \mid x) \,\|\, \pi_\theta(\cdot \mid x)\right)\right]$$

MLE Equals KL Minimization

- Start from expected log-likelihood: $\mathbb{E}_{y\sim p}\left[\log \pi_\theta(y)\right]$

- Add and subtract $\log p(y)$

$$\mathbb{E}_p\left[\log \pi_\theta(y)\right] = \mathbb{E}_p\left[\log p(y)\right] - \mathbb{E}_p\left[\log \frac{p(y)}{\pi_\theta(y)}\right]$$

- Therefore:

$$\mathbb{E}_p\left[\log \pi_\theta(y)\right] = -H(p) \; - \; \mathrm{KL}(p \parallel \pi_\theta)$$

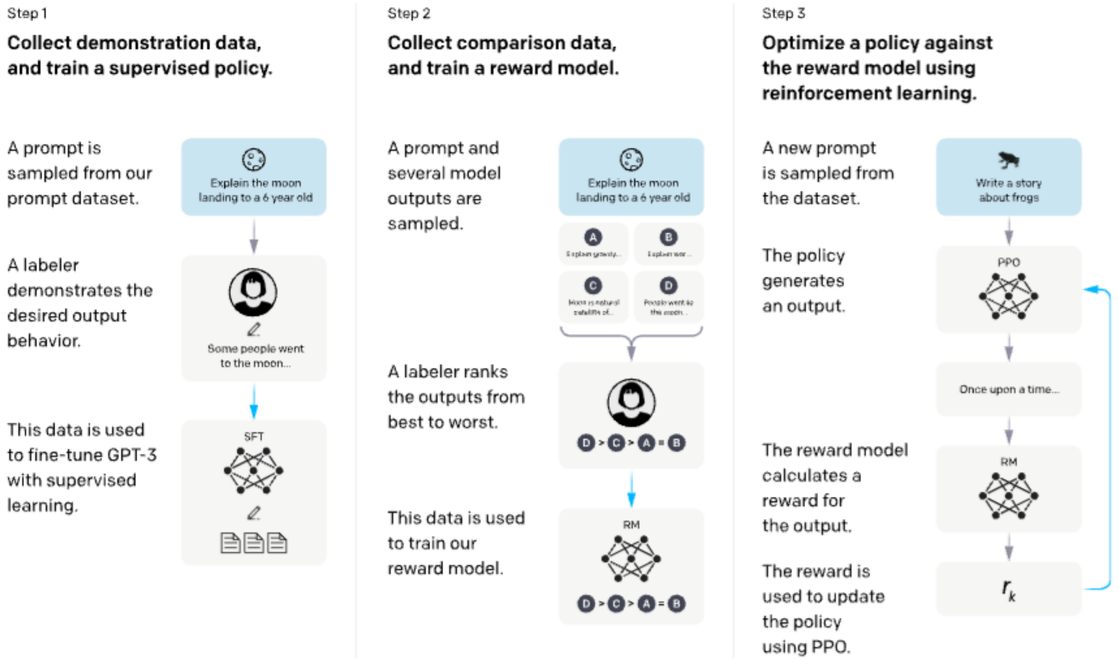
Why SFT Defines π_{ref}

- SFT gives a competent instruction-following baseline
- Later optimization updates π_θ
- π_θ but keeps it close to π_{ref}
- This prevents:
  - **language drift** (losing fluency / formatting)
  - **reward exploitation** (overfitting the proxy)
  - **instability** (large distribution shift)

$$\pi_{\text{ref}} \quad := \quad \text{SFT checkpoint}$$

InstructGPT Pipeline

- **Step 1:** Supervised Fine-Tuning (SFT) → produces π_{ref}
- **Step 2:** Preference rankings → train Reward Model $r_\phi(x, y)$
- **Step 3:** Optimize π_θ against r_ϕ with **KL control** (PPO)



KL-Regularized Post-Training Objective

- Reward signal $r(x, y)$ (from RM / verifier / rubric)
- Reference policy $\pi_{\text{ref}}(\cdot \mid x)$ (usually SFT)
- Optimize:

$$\max_{\theta} \; \mathbb{E}_{y \sim \pi_\theta(\cdot|x)}[\, r(x, y)\,] \; - \; \beta \, \mathbb{E}_{y \sim \pi_\theta(\cdot|x)} \left[\log \frac{\pi_\theta(y \mid x)}{\pi_{\text{ref}}(y \mid x)} \right]$$

Equivalent Constrained Form (Trust Region View)

- Penalized form:

$$\max_{\theta} \; \mathbb{E}_{y\sim\pi_\theta}[r(x,y)] - \beta\, \mathbb{E}_{y\sim\pi_\theta}\left[\log \frac{\pi_\theta(y \mid x)}{\pi_{\text{ref}}(y \mid x)}\right]$$

- Constrained form (same spirit):

$$\max_{\theta} \; \mathbb{E}_{y\sim\pi_\theta}[r(x,y)] \quad \text{s.t.} \quad \mathbb{E}_x\left[\mathrm{KL}(\pi_\theta(\cdot \mid x) \,\|\, \pi_{\text{ref}}(\cdot \mid x))\right] \le \varepsilon$$

- Intuition: “Improve reward, but stay in a KL ball around π_{ref}

Proof: Closed-Form Optimal Policy 1/2

Fix a prompt x. Optimize over distributions $\pi(\cdot \mid x)$ on a finite set of completions $y \in \mathcal{Y}$

$$\max_{\pi} \sum_{y \in \mathcal{Y}} \pi(y \mid x)\, r(x, y) \; - \; \beta \sum_{y \in \mathcal{Y}} \pi(y \mid x) \log \frac{\pi(y \mid x)}{\pi_{\text{ref}}(y \mid x)}$$

subject to (for every prompt): $\sum_{y \in \mathcal{Y}} \pi(y \mid x) = 1$ and $\pi(y \mid x) \geq 0$

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Lagrangian: $\mathcal{L}(\pi, \lambda) = \sum_{y} \pi(y \mid x)\, r(x, y) - \beta \sum_{y} \pi(y \mid x) \log \frac{\pi(y \mid x)}{\pi_{\text{ref}}(y \mid x)} + \lambda \left(\sum_{y} \pi(y \mid x) - 1 \right).$

Proof: Closed-Form Optimal Policy 1/2

Fix a prompt x. Optimize over distributions $\pi(\cdot \mid x)$ on a finite set of completions $y \in \mathcal{Y}$

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Stationary condition (for each y)

$$\frac{\partial \mathcal{L}}{\partial \pi(y \mid x)} = r(x, y) - \beta \left(\log \frac{\pi(y \mid x)}{\pi_{\text{ref}}(y \mid x)} + 1 \right) + \lambda = 0.$$

Proof: Closed-Form Optimal Policy 2/2

Rearrange:

$$\log \frac{\pi(y \mid x)}{\pi_{\text{ref}}(y \mid x)} = \frac{r(x, y)}{\beta} + \frac{\lambda}{\beta} - 1.$$

Exponentiate:

$$\pi(y \mid x) = \pi_{\text{ref}}(y \mid x) \exp\left(\frac{r(x, y)}{\beta}\right) \cdot \exp\left(\frac{\lambda}{\beta} - 1\right).$$

Let $Z(x) := \sum_y \pi_{\text{ref}}(y \mid x) \exp(r(x, y)/\beta)$. Enforcing normalization gives

$$\exp\left(\frac{\lambda}{\beta} - 1\right) = \frac{1}{Z(x)}.$$

Conclusion

$$\pi^*(y \mid x) = \frac{\pi_{\text{ref}}(y \mid x) \exp(r(x, y)/\beta)}{\sum_{y'} \pi_{\text{ref}}(y' \mid x) \exp(r(x, y')/\beta)}$$

Reference = Imitation Prior; Reward = Tilt

- Closed-form optimum:

$$\pi^*(y \mid x) = \frac{\pi_{\text{ref}}(y \mid x)\exp\left(r(x,y)/\beta\right)}{Z(x)}$$

- Log form (useful intuition):

$$\log \pi^*(y \mid x) = \log \pi_{ref}(y \mid x) + \frac{r(x,y)}{\beta} - \log Z(x)$$

- Effect of β
  - $\beta \downarrow$: stronger tilt toward high reward (more drift)
  - $\beta \uparrow$: weaker tilt (stay close to π_{ref})

Token-Level Vs Sequence-Level KL

- Sequence KL (per prompt)

$$\mathrm{KL}\big(\pi_\theta(\cdot \mid x) \,\|\, \pi_{\text{ref}}(\cdot \mid x)\big) = \mathbb{E}_{y\sim\pi_\theta(\cdot|x)}\left[\log \frac{\pi_\theta(y \mid x)}{\pi_{\text{ref}}(y \mid x)}\right]$$

- Autoregressive factorization:

$$\log \frac{\pi_\theta(y \mid x)}{\pi_{\text{ref}}(y \mid x)} = \sum_{t=1}^{T} \log \frac{\pi_\theta(y_t \mid x, y_{<t})}{\pi_{\text{ref}}(y_t \mid x, y_{<t})}$$

- Therefore (practical lens):

$$\mathrm{KL}(\text{sequence}) = \sum_{t=1}^{T} \mathbb{E}_{y\sim\pi_\theta}\left[\log \frac{\pi_\theta(y_t \mid x, y_{<t})}{\pi_{\text{ref}}(y_t \mid x, y_{<t})}\right]$$

(expectation is under rollouts from π_θ).

Why KL Is Non-Negotiable

- Post-training uses proxy rewards (RM / heuristics / verifiers) → imperfect signals
- Without a KL anchor, optimization can exploit proxy errors:

$$\max_{\theta} \ \mathbb{E}_{y\sim\pi_\theta}[r(x,y)] \quad \Rightarrow \quad \text{seek } y \text{ that hacks } r$$

- KL penalty enforces conservative updates:

$$\max_{\theta} \ \mathbb{E}[r] \ - \ \beta\,\mathbb{E}\left[\log \frac{\pi_\theta}{\pi_{\text{ref}}}\right]$$

- Typical failure modes when KL is too weak:
  - **Reward hacking** (proxy exploitation)
  - **Distribution shift** (off-policy RM errors worsen)
  - **Language/style drift** (verbosity, weird formatting, repetition)

Lecture B

PPO Motivation: Unconstrained Updates Can Blow Up

- Objective we would like to improve (sequence-level):

$$J(\theta) = \mathbb{E}_{y \sim \pi_\theta(\cdot|x)}[R(x, y)]$$

- First-order “surrogate” around π_{old} using importance sampling:

$$J(\theta) \approx \mathbb{E}_{y \sim \pi_{\text{old}}} \left[\frac{\pi_\theta(y \mid x)}{\pi_{\text{old}}(y \mid x)} R(x, y) \right]$$

- Problem: the ratio $\pi_\theta / \pi_{\text{old}}$ can become large $\Rightarrow$ unstable jumps

- Fix: enforce conservative updates (trust-region / clipping)

Importance Ratio For Token Actions

- State/action for an autoregressive LM: $s_t = (x, y_{<t}), \qquad a_t = y_t$

- Token-level importance ratio:

$$r_t(\theta) = \frac{\pi_\theta(a_t \mid s_t)}{\pi_{\text{old}}(a_t \mid s_t)} = \frac{\pi_\theta(y_t \mid x, y_{<t})}{\pi_{\text{old}}(y_t \mid x, y_{<t})}$$

- Sequence-level ratio factorizes:

$$\frac{\pi_\theta(y \mid x)}{\pi_{\text{old}}(y \mid x)} = \prod_{t=1}^{T} r_t(\theta), \qquad \log \frac{\pi_\theta(y \mid x)}{\pi_{\text{old}}(y \mid x)} = \sum_{t=1}^{T} \log r_t(\theta)$$

PPO Clipped Surrogate Objective

- Define the clipped ratio:

$$\text{clip}(r_t, 1-\epsilon, 1+\epsilon)$$

- PPO surrogate (per token):

$$L^{\text{clip}}(\theta) = \mathbb{E}\left[\min\left(r_t(\theta)A_t,\ \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)A_t\right)\right]$$

- Total objective sums over tokens (and batch):

$$\max_{\theta}\ \mathbb{E}\left[\sum_{t=1}^{T} L_t^{\text{clip}}(\theta)\right]$$

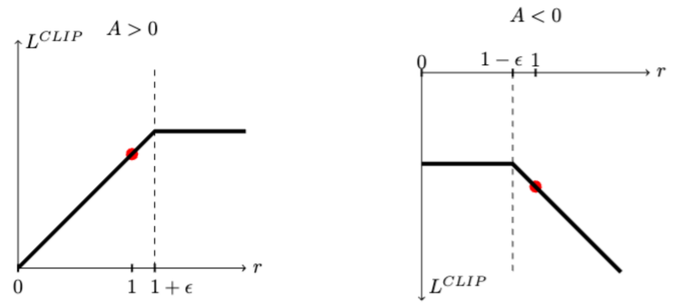


Figure 1: Plots showing one term (i.e., a single timestep) of the surrogate function L^{CLIP} as a function of the probability ratio r, for positive advantages (left) and negative advantages (right). The red circle on each plot shows the starting point for the optimization, i.e., $r = 1$. Note that L^{CLIP} sums many of these terms.

RLHF Reward: Score Minus KL

- Reward model score (from Lecture 7):

$$r_{\mathrm{RM}}(x, y) \in \mathbb{R}$$

- KL penalty to reference policy:

$$\mathrm{KL}\big(\pi_\theta(\cdot \mid x) \,\|\, \pi_{\mathrm{ref}}(\cdot \mid x)\big) = \mathbb{E}_{y \sim \pi_\theta}\left[\log \frac{\pi_\theta(y \mid x)}{\pi_{\mathrm{ref}}(y \mid x)}\right]$$

- Define the RLHF return:

$$R(x, y) = r_{\mathrm{RM}}(x, y) \; - \; \beta \; \mathrm{KL}\big(\pi_\theta(\cdot \mid x) \,\|\, \pi_{\mathrm{ref}}(\cdot \mid x)\big)$$

Advantage Estimation In RLHF

- Sequence return (single scalar):

$$R(x, y) = r_{\mathrm{RM}}(x, y) - \beta\, \mathrm{KL}(\pi_\theta \| \pi_{\mathrm{ref}})$$

- Token-level credit via advantage:

$$A_t = \hat{Q}(s_t, a_t) - \hat{V}(s_t)$$

- In the simplest sequence-level setting (common in LLM RLHF):

$$\hat{Q}(s_t, a_t) \approx R(x, y), \qquad A_t \approx R(x, y) - b(x)$$

where $b(x)$ is a baseline (e.g., value model or mean return for prompt).

PPO: What Can Go Wrong

- **Instability**: high-variance returns + on-policy sampling → noisy gradients
- **Over-optimization of proxy**: RM exploitation; quality may degrade
- **KL collapse / explosion**: too small β → drift; too large β → no learning
- **Engineering cost**: rollouts + careful tuning

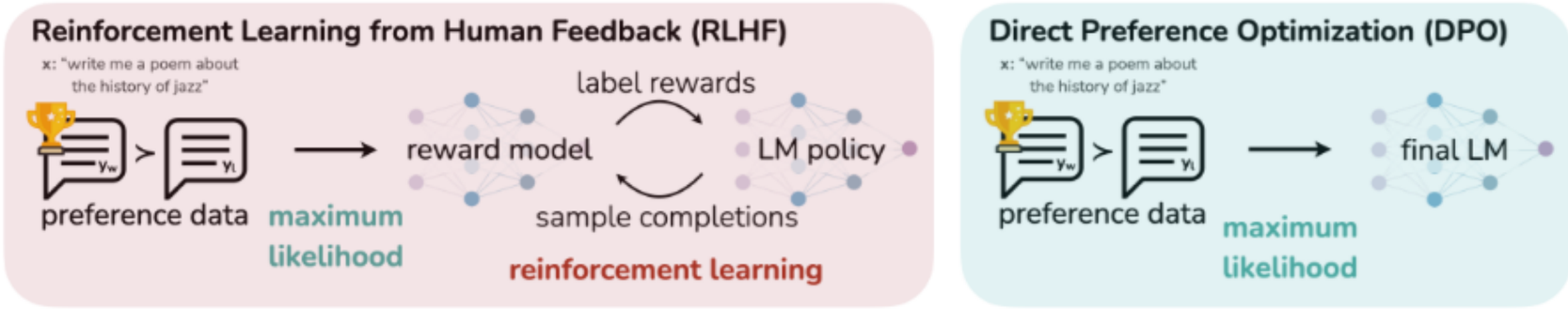
Direct Prference Optimization (DPO)

DPO: Optimize From Preference Pairs

- Data: $(x,\ y^+,\ y^-),\ y^+ \succ y^-$
- Choose a fixed reference π_{ref} (usually SFT)
- Train π_θ, by maximizing a preference log-likelihood

$$\max_\theta \ \mathbb{E}\left[\log P_\theta(y^+ \succ y^- \mid x)\right]$$

- Key move: parameterize $P_\theta(\cdot)$ using **policy log-prob ratios** (next slides)



Key Identity From The KL-Regularized Optimum

- From the previous theorem, for fixed x

$$\pi^*(y \mid x) \propto \pi_{\text{ref}}(y \mid x)\exp\Big(r(x,y)/\beta\Big)$$

- Taking logs and rearranging gives:

$$r(x,y) = \beta\Big(\log \pi^*(y \mid x) - \log \pi_{\text{ref}}(y \mid x)\Big) + C(x)$$

where

$$C(x) = \beta \log Z(x) \quad \text{and} \quad Z(x) = \sum_{y'} \pi_{\text{ref}}(y' \mid x)\exp(r(x,y')/\beta)\,.$$

Bradley–Terry Likelihood In Policy Log-Ratios

- Bradley–Terry model (Lecture 7):

$$P(y^+ \succ y^- \mid x) = \sigma\Big(r(x, y^+) - r(x, y^-)\Big)$$

Substitute the difference form (dropping $C(x)$) : $r(x, y^+) - r(x, y^-) = \beta\Big(\Delta_\theta - \Delta_{\text{ref}}\Big)$

where

$$\Delta_\theta = \log \pi_\theta(y^+ \mid x) - \log \pi_\theta(y^- \mid x), \qquad \Delta_{\text{ref}} = \log \pi_{\text{ref}}(y^+ \mid x) - \log \pi_{\text{ref}}(y^- \mid x)\,.$$

- Therefore:

$$\boxed{P_\theta(y^+ \succ y^- \mid x) = \sigma\Big(\beta(\Delta_\theta - \Delta_{\text{ref}})\Big)}$$

DPO Loss

- Given preference pairs (x, y^+, y^-), define

$$u_\theta(x, y^+, y^-) = \beta\Big[(\log \pi_\theta(y^+ \mid x) - \log \pi_\theta(y^- \mid x)) - (\log \pi_{\text{ref}}(y^+ \mid x) - \log \pi_{\text{ref}}(y^- \mid x))\Big].$$

- Then the DPO objective is maximum likelihood:

$$\max_\theta \ \mathbb{E}\Big[\log \sigma(u_\theta)\Big]$$

- Equivalently, minimize the loss:

$$\mathscr{L}_{\text{DPO}}(\theta) = -\,\mathbb{E}_{(x,y^+,y^-)}\Big[\log \sigma(u_\theta(x, y^+, y^-))\Big]$$

MLE On Preferences ⇒ DPO Loss

- Model:

$$P_\theta(y^+ \succ y^- \mid x) = \sigma(u_\theta(x, y^+, y^-))$$

- Log-likelihood over dataset $\mathscr{D}$:

$$\ell(\theta) = \sum_{(x,y^+,y^-)\in\mathscr{D}} \log \sigma(u_\theta(x, y^+, y^-))$$

- Negative log-likelihood:

$$-\ell(\theta) = \sum_{(x,y^+,y^-)\in\mathscr{D}} -\log \sigma(u_\theta(x, y^+, y^-)) \;\equiv\; |\mathscr{D}| \cdot \mathscr{L}_{\text{DPO}}(\theta)$$

What DPO Is Doing (Gradient View)

- For one pair, loss is $\ell(\theta) = -\log \sigma(u_\theta)$. Then,

$$\frac{\partial \ell}{\partial u_\theta} = -\left(1 - \sigma(u_\theta)\right).$$

And,

$$\frac{\partial u_\theta}{\partial \theta} = \beta\Big(\nabla_\theta \log \pi_\theta(y^+ \mid x) - \nabla_\theta \log \pi_\theta(y^- \mid x)\Big).$$

So

$$\nabla_\theta \ell(\theta) = -\beta(1 - \sigma(u_\theta))\Big(\nabla_\theta \log \pi_\theta(y^+ \mid x) - \nabla_\theta \log \pi_\theta(y^- \mid x)\Big).$$

Role Of β in DPO

- Appears in the preference model:

$$P_\theta(y^+ \succ y^- \mid x) = \sigma\Big(\beta(\Delta_\theta - \Delta_{\text{ref}})\Big)$$

- Appears in gradient magnitude:

$$\|\nabla_\theta \ell\| \;\propto\; \beta\,(1 - \sigma(u_\theta))$$

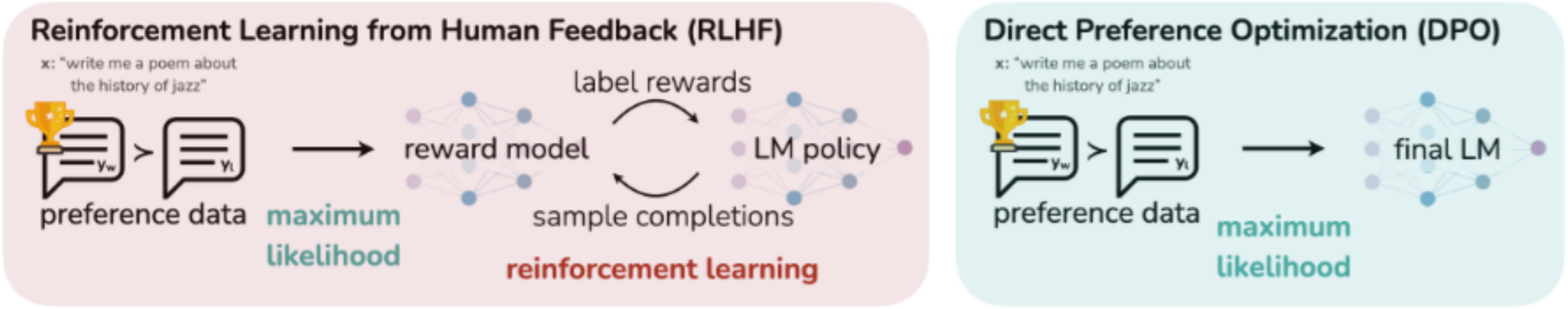
- Connection to KL-regularized optimum:

$$\pi^*(y \mid x) \propto \pi_{\text{ref}}(y \mid x)\exp(r(x, y)/\beta),$$

so β controls how strongly reward reshapes the reference distribution.

DPO Pipeline

- Inputs: preference pairs (x, y^+, y^-)
- Compute: $\log \pi_\theta(y^\pm \mid x)$ and $\log \pi_{\text{ref}}(y^\pm \mid x)$
- Optimize: $\mathscr{L}_{\text{DPO}}(\theta) = -\log \sigma(u_\theta)$



DPO Practical Checklist (What To Monitor)

- Monitor preference accuracy / win-rate on held-out pairs:

$$\mathbb{P}\left(u_\theta(x, y^+, y^-) > 0\right)$$

- Monitor KL drift from reference (token KL estimate):

$$\widehat{\mathrm{KL}} \approx \frac{1}{T} \sum_t \left(\log \pi_\theta - \log \pi_{\mathrm{ref}}\right)$$

- Watch length/verbosity drift (often correlates with log-prob shifts)

- Reference choice: π_{ref} fixed SFT vs moving reference (changes the target)